

Experiment Design for Computer Sciences (0AL0400)

Report 1 Review

Claus Aranha

caranha@cs.tsukuba.ac.jp

University of Tsukuba, Department of Computer Sciences

June 7, 2024

Version 2024

Outline

- General Comments about the Reports
- Specific Comments about the reports

Talking about your experiment: Know your audience

- Three paragraphs of introduction before explaining the experience. What does your reader want to know?
- The objective of each text is different. There is no size fits all solution.

Interpreting results honestly

- Interpret the results honestly. Do not say the results are clear when they are not.
- Poke holes in your experiment, and do curious questioning.
- Choose experiments that are important for you for some reason (you need the result for some reason). If the results are important for you, then you will question them more honestly.
- (Note: The report is important to you $\neq$ the result is important to you)

Why Kaggle is not a good object of analysis

- Kaggle data is made for testing algorithms. The data itself is not important for you (usually)
- Kaggle data is made for testing algorithm: It was not gathered for analysis of the data itself
- Little information about how it was gathered, possible issues.
- Possible issues also change depending on the question you want to ask.

Making Good questions

- Common question: "Does X has an influence on Y?" The answer is very often yes. This is not a good question.
 - Example: Estimation of Pi by Monte Carlo: Does N have an effect? Of course it does. It is obvious.
 - A more interesting question: How big N has to be for me to get a "good enough" PI for some reason?
- "How much of an influence X has on Y?"; "Does X has enough of an influence on Y that we should care about X?"

Simulated Experiments

- How much can we learn from simulated experiments? We can simulate whatever we want.
- How do we validate results from simulated experiments?

Specific Comments

Adversarial Neural Networks

Experiment on Adversarial Neural Network: Does an adversarial image generated on network A affects the result of network B?

Accuracy on Network A: 33% (original accuracy is not reported).

Accuracy on network B: 90% (original accuracy is 92%).

How would you interpret this result? Noise is not considered correctly. What possible sources of noise could be in this experiment, and how would you evaluate them?

Specific Comments

Comparison on programming languages

- Comparison of Python, Java and C on quicksort. C was much lower. Why is that?
- Importance of implementation in programming experiments

Specific Comments

Learning English

- One experiment about learning English; (Memorizing words)
- No problem with the experiment. But to learn English, you should focus on Output, not Input;

Specific Comments

Measuring time for Caramel

- Measuring state change with time cooking.
- Experiment found good transitions.
- Idea for follow up: If 130s seems to be the optimal heating time, how reproducible is this? Does this always work?

Specific Comments

Simulated Experiments

- Comparing Simulated Data vs Real Birth data. How similar are the results?
- Green light experiment: How to validate premises?

Specific Comments

Baumkuchen Experiment

- Options: Two different samples or two of the same sample
- The importance of controls!

Specific Comments

Music and Study

- A few "self experiments"
- How do you define a control in this case, is it even necessary?
- Is it necessary to test on other people? Depend on what you care about!